Purret

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Project Overview

Purret is a WiFi-enabled laser turret designed for interactive cat play, combining robotics, video streaming, and a gamified interface for remote laser control. The system is built around four modular components – Hardware Control, Video Streaming, Master API, and Frontend – each running in its own Docker container. A wireguard tunnel is added ontop to allow for remote access.

Achievements & Metrics

Nick: Team lead, Video Container, and Setup.

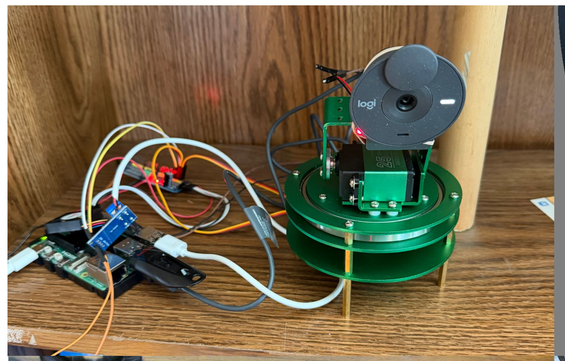
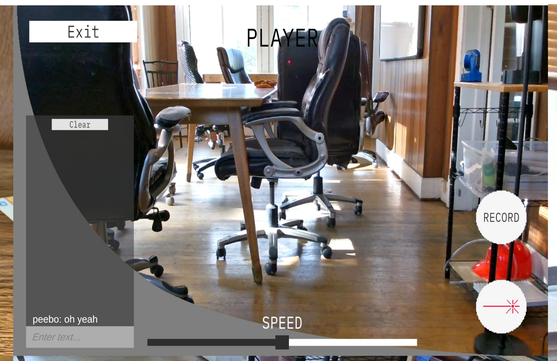
Mark: Created the master api that communicated between turret and frontend.

Tarik: Designed and implemented frontend Unity project.

Keith: Constructed the turret and made an api to access controls for the turret.

Goals

- Hardware Control: Laser/servo controls equipped with safety features
- Video and Streaming Functional: Live Wifi streaming, recording, & health monitoring
- Core Backend/API Finalized: API with secure single-user control and error handling
- Frontend Fully Integrated: Real time monitoring and full control
- User Experience Enhanced: Gamified the experience with a chat for spectators

Technical Stack

Hardware & Video Control	Python (subprocess, serial)
API Services	Python, FastAPI
Media Server	MediaMTX
Video Processing	Ffmpeg
Frontend	Unity (C#)
Containerization	Docker
Networking	WireGuard (remote access)
Hardware Platform	Raspberry Pi